

Algebra II - Tutorial 3 - Date: 2/2/2025

1. Let $G = S_4$ and G' the symmetric group on the three elements of S_4 , $\Pi_1 = (12)(34)$, $\Pi_2 = (13)(24)$ and $\Pi_3 = (14)(23)$. Define $\phi : G \rightarrow G'$ as follows: $\phi(\sigma)$ is the permutation on $\{\Pi_1, \Pi_2, \Pi_3\}$. Determine the kernel of ϕ . What are all the subgroups which contain K .
2. Let $G = \langle x \rangle$ be a cyclic group of order 12 and $G' = \langle y \rangle$ a group of order 6. Let $\phi : G \rightarrow G'$ be given by $\phi(x^i) = y^i$. Exhibit the correspondence referred to in the correspondence theorem explicitly.
3. Let G be a group of order p^2 . Show that it has at least one subgroup of order p and if it has a unique subgroup of order p then it is a cyclic group.
4. Prove or disprove: Every subgroup of index 2 is a normal subgroup.
5. Prove or disprove: $\mathbb{Z}_p \times \mathbb{Z}_p$ is a cyclic group.
6. Let $|G| = p^n$ where p is a prime. Does G contain an element of order p ?
7. Does a group of order 35 contain an element of order 5? of order 7?
8. Let H and K be subgroups of a group G .
 - (a) Prove that the intersection $xH \cap yK$ of two cosets of H and K is either empty or else is a coset of the subgroup $H \cap K$.
 - (b) Prove that if H and K have finite index in G then $H \cap K$ also has finite index in G .
9. In each of the following cases, determine whether or not G is isomorphic to the product group $H \times K$.
 - (a) $G = \mathbb{R}^\times$, $H = \{\pm 1\}$, $K = \{\text{positive real numbers}\}$.
 - (b) $G = \mathbb{C}^\times$, $H = \{\text{unit circle}\}$, $K = \{\text{positive real numbers}\}$.
 - (c) $G = \{\text{invertible upper triangular } 2 \times 2 \text{ matrices}\}$, $H = \{\text{invertible diagonal matrices}\}$, $K = \{\text{upper triangular matrices with diagonal entries } 1\}$.
10. Classify groups of order 6 by analyzing the following three cases:
 - (a) G contains an element of order 6.
 - (b) G contains an element of order 3 but none of order 6.
 - (c) All elements of G have order 1 or 2.